Probabilistic Machine Learning

Lecture 03: Generative Models and Naive Bayes

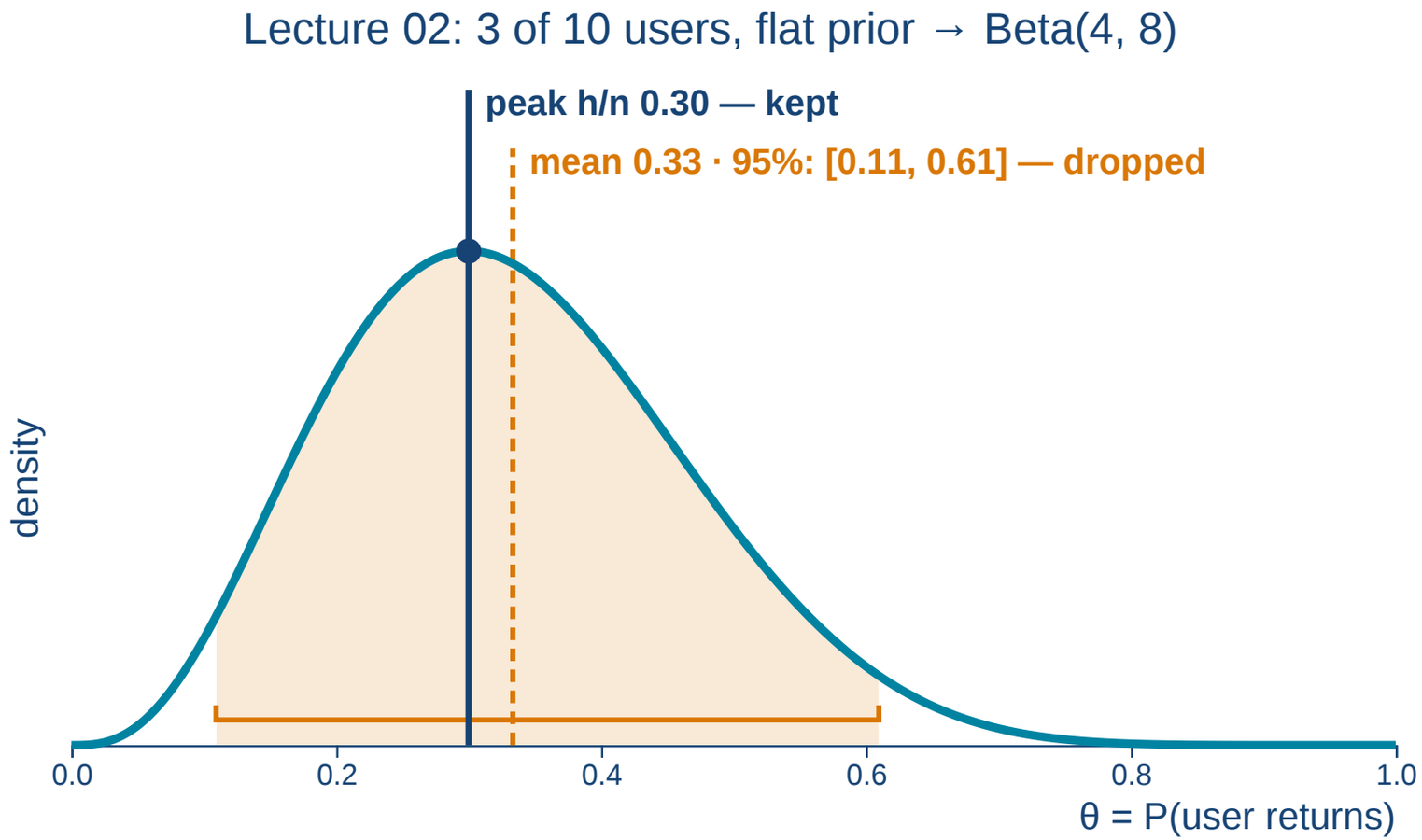
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Speaker notes

Three modules. Keep the peak: maximum likelihood (MLE) and maximum a posteriori (MAP) estimation on the Beta–Bernoulli model of Lecture 02, where the posterior can be drawn and therefore the information a point estimate discards (the spread) is visible. The generative route: a classifier obtained from Bayes' theorem by modelling $P(y)$ and $p(x \mid y)$, and the exponential number of parameters a full $p(x \mid y)$ requires. Naive Bayes: conditional independence of the attributes given the class, which reduces that number from $2^p - 1$ to p per class for p binary attributes.

Every step of naive Bayes is an instance of earlier material: fitting is the MLE of one categorical per attribute and class, Laplace smoothing is the posterior mean under a flat Beta (or Dirichlet) prior, and correlated attributes are counted twice, which makes the predicted probabilities over-confident. Fraud detection (base rate 1 in 1,000, binary red flags) is the running example. Lecture 04 treats linear and logistic regression as models of a conditional distribution and compares logistic regression with naive Bayes.

Where We Left Off



One posterior, three readings:

- **peak** $0.30 = h/n$
- **mean** 0.33
- **95% interval** $[0.11, 0.61]$

Today keeps only the **peak** — and then asks what a classifier built on probability looks like.

Speaker notes

The posterior of Lecture 02: 3 of 10 users return, flat prior Beta(1, 1), so the posterior is Beta(1 + 3, 1 + 7) = Beta(4, 8). Its three summaries: the mode $(a-1)/(a+b-2) = 3/10 = 0.30$, which equals h/n because the prior is flat; the mean $a/(a+b) = 4/12 = 0.33$; the central 95% interval $[0.109, 0.610]$ from the Beta quantiles. The standard deviation is 0.13.

The peak (navy in the plot) is the only one of the three that can be found without the normaliser, by maximising; mean and interval (amber) need the whole distribution. Fitting by least squares or by cross-entropy is exactly this kind of peak: least squares is the MLE under Gaussian noise, cross-entropy is the negative log-likelihood of a Bernoulli or categorical output, and adding weight decay turns either into a MAP estimate (Lecture 04).

Today: Keep the Peak, Assume It Apart

Lecture 01's table, rows 1 and 2 — and the first classifier built straight from probability:

shortcut	instead of the ideal	what is lost	where
Keep the peak	the most probable parameter — MLE or MAP	how sure we are about it	a coin · every count
Assume it apart	factorise the joint by independence	the dependences assumed away	naive Bayes

Keep the peak
on a coin

→

The generative route
a classifier from Bayes

→

Naive Bayes
assume it apart

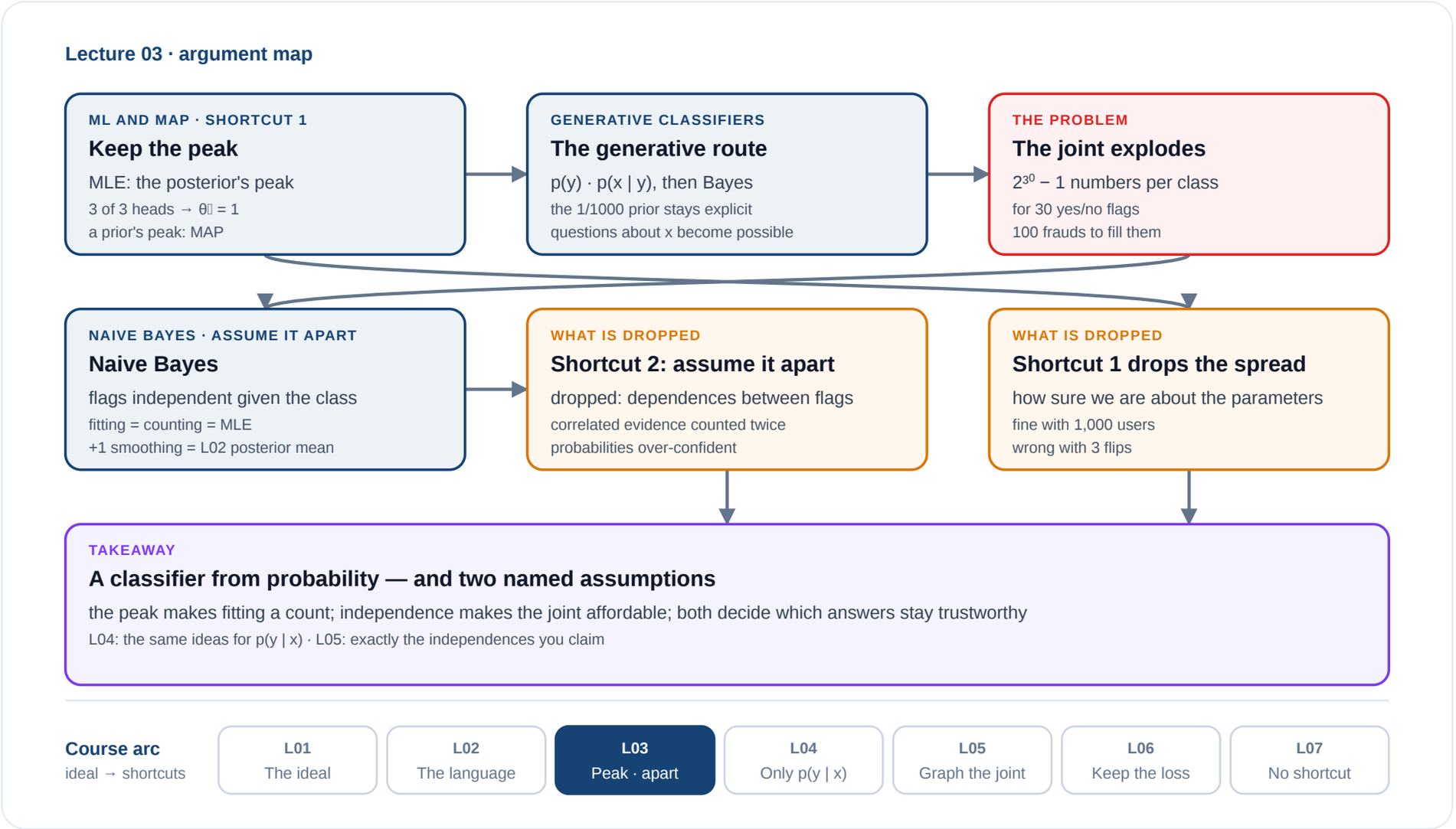
Speaker notes

The shortcut table of Lecture 01 lists the approximations the course makes to the ideal of carrying full distributions. This lecture covers the first two. Keep the peak replaces the posterior over a parameter by its most probable value (MLE with a flat prior, MAP otherwise) and discards the uncertainty about that value. Assume it apart replaces a joint distribution by a product of smaller factors under an independence assumption and discards the dependences between the factors.

Naive Bayes takes both: each class-conditional factor $p(x_j \mid y)$ is estimated by its peak (a relative frequency), and the joint $p(x \mid y)$ is replaced by the product of these factors. A generative classifier as such requires neither shortcut: independence is what makes the joint affordable, and the peak is what turns fitting into counting.

Running example: fraud detection with a base rate of 1 in 1,000 (prior odds 1 : 999) and a handful of binary red flags.

Argument Map



Speaker notes

The map links three modules, one problem and two costs. Keep the peak: the MLE is the posterior's peak (3 heads in 3 flips give $\hat{\theta} = 1$), MAP is the peak with a prior; it drops the spread, which is harmless with 1,000 observations and misleading with 3. The generative route models $p(y)$ and $p(x \mid y)$ and converts with Bayes, keeping the 1/1,000 prior explicit; its problem is the joint, $2^{30} - 1$ numbers per class for 30 binary flags against 100 observed frauds. Naive Bayes assumes the flags independent given the class, so fitting becomes counting (an MLE) and +1 smoothing becomes the posterior mean of Lecture 02; it drops the dependences between flags, so correlated evidence is counted twice and the probabilities are over-confident. The course-arc strip places the lecture after the language of probability (L02) and before models of $p(y \mid x)$ only (L04) and graphical models (L05), which state exactly which independences are assumed.

Maximum Likelihood and MAP

Guiding Question: Last week's posterior gave a peak, a mean and an interval. What do we lose when we keep only the peak — and why does almost everyone do it?

The Likelihood, Read Along the Other Axis

The **likelihood** is the probability of the observed data, read as a function of the **parameter** with the data held fixed: $L(\theta) = p(D \mid \theta)$. The **maximum likelihood estimate** is its peak, $\hat{\theta}_{\mathrm{ML}} = \arg\max_\theta p(D \mid \theta)$.

Lecture 02, 3 of 10 users

$$p(\theta \mid D) \propto \theta^3 (1-\theta)^7 \cdot 1$$

With a **flat prior** the posterior *is* the normalised likelihood — so its peak $h/n = 0.3$ is the maximum likelihood estimate.

Not a distribution over θ

$$\int_0^1 \theta^3 (1-\theta)^7 \mathrm{d}\theta = \frac{1}{1320}$$

The likelihood is a probability over **data**. Read along θ it need not integrate to 1 — only the posterior does.

Speaker notes

With a flat prior $p(\theta) = 1$ on $[0, 1]$, Bayes' theorem gives $p(\theta \mid D) = p(D \mid \theta)/p(D)$: the posterior is the likelihood divided by a constant, so both have the same shape and the same arg max. The MLE is therefore the MAP under a flat prior. With any other prior the two peaks differ.

The normaliser here is $\int_0^1 \theta^3(1-\theta)^7\, d\theta = B(4, 8) = 3!\,7!/11! = 1/1320$, the Beta function that turns the likelihood into the Beta(4, 8) density. That the integral is not 1 shows that the likelihood is not a density over θ: it is a probability of the data for each fixed θ, and summed over all possible datasets it gives 1, not integrated over θ. A common error is to read "the likelihood of $\theta = 0.3$" as "the probability that $\theta = 0.3$"; only the posterior supports the second statement, and only after a prior has been chosen. The likelihood is also defined only up to a constant factor in θ (for example the binomial coefficient $\binom{10}{3}$ when only the count is recorded), which changes neither its peak nor the posterior.

Deriving h/n Without a Posterior

The i.i.d. assumption of Lecture 02 lets the likelihood factorise over rows; the logarithm turns the product into a sum without moving the peak:

$$\ell(\theta) = \log \prod_{i=1}^{n} \theta^{y_i} (1-\theta)^{1-y_i} = h \log \theta + (n-h) \log(1-\theta)$$

$$\frac{d\ell}{d\theta} = \frac{h}{\theta} - \frac{n-h}{1-\theta} = 0 \quad \Longrightarrow \quad \hat{\theta}_{\mathrm{ML}} = \frac{h}{n}$$

> Last week's peak, found by **setting a derivative to zero —** no prior, no normaliser, no integral. The recipe works unchanged for a million parameters, where nobody can draw the posterior.

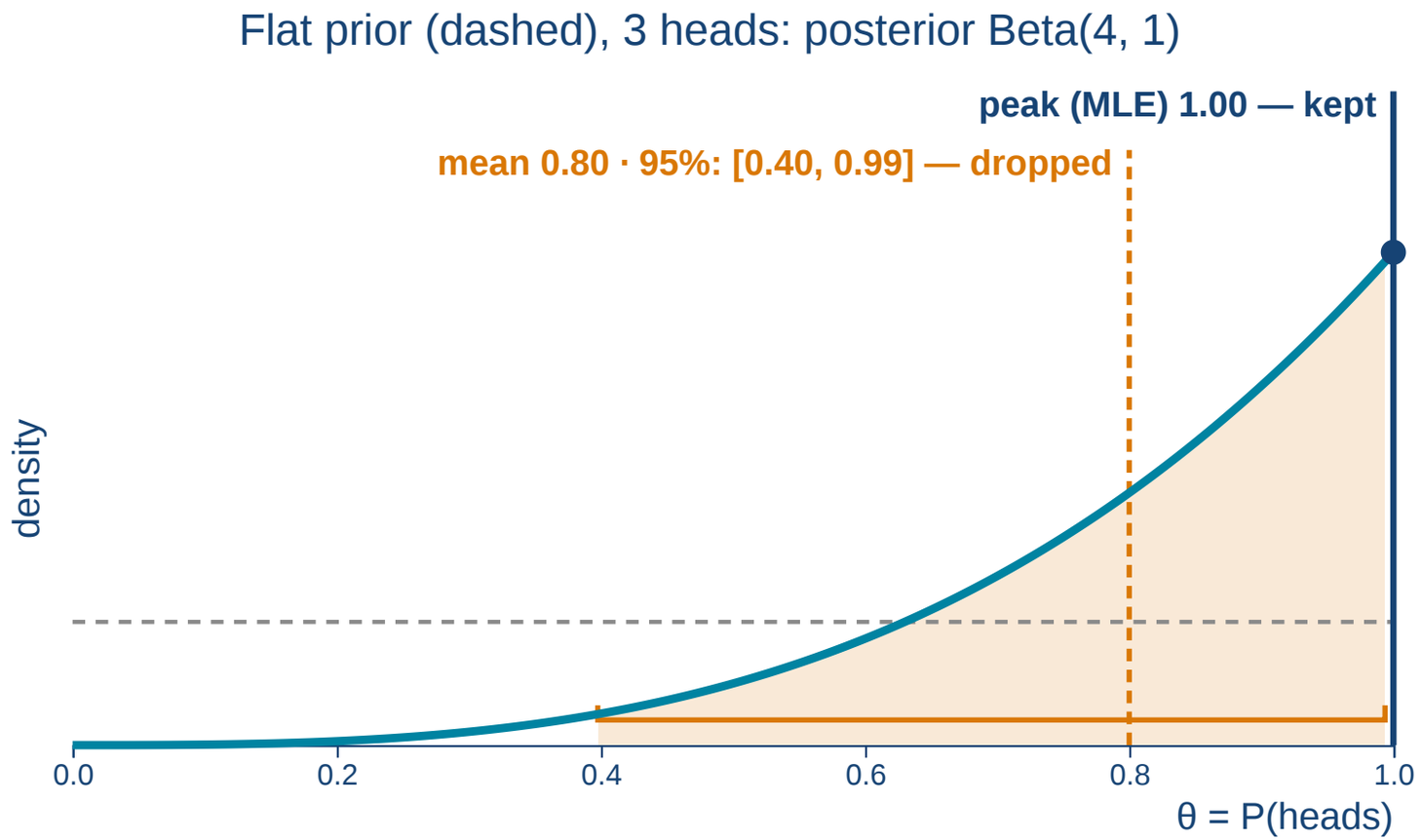
Speaker notes

Steps of the derivation. Under the i.i.d. assumption $p(D \mid \theta) = \prod_i \theta^{y_i}(1-\theta)^{1-y_i} = \theta^h(1-\theta)^{n-h}$ with $h = \sum_i y_i$, so the data enter only through the count h (a sufficient statistic). Taking the logarithm is allowed because $\log$ is strictly increasing, so it does not move the arg max; it also avoids numerical underflow of a product of many small probabilities and makes the derivative a sum of per-row terms. Solving $h/\theta = (n-h)/(1-\theta)$ gives $h(1-\theta) = (n-h)\theta$, i.e. $\theta = h/n$. The second derivative $-h/\theta^2 - (n-h)/(1-\theta)^2$ is negative, so this is a maximum. For $h = 0$ or $h = n$ the maximum lies on the boundary ($\theta = 0$ or 1), where the derivative is not zero; the formula h/n still holds.

The reason the peak is used in practice: it replaces integration (the normaliser $p(D)$) by optimisation. For models with many parameters there is no closed form, and the maximum is found numerically, typically by gradient methods on the negative log-likelihood.

The Peak Can Lie

Three flips, **three heads**. Maximum likelihood: $\hat{\theta}_{\mathrm{ML}} = 3/3 = 1$.



Predict before the second click: where do the posterior mean and the 95% interval lie?

Speaker notes

Step 1 marks the peak of Beta(4, 1) at $\theta = 1$: the MLE $3/3$ assigns probability 0 to tails, a claim no finite number of heads can justify. Step 2 adds the posterior mean and the central 95% interval. Beta(4, 1) has density $4\theta^3$ and CDF θ^4, so the mean is $4/5 = 0.8$ and the 2.5% and 97.5% quantiles are $0.025^{1/4} = 0.398$ and $0.975^{1/4} = 0.994$. The posterior therefore says "probably above one half, possibly well below 1", whereas the peak says "never tails".

The data and the model are not at fault; the information loss comes from reducing a skewed posterior to its mode. For a posterior with its mode on the boundary of the parameter space, the mode is an extreme point of the distribution and a poor summary. The same effect appears in naive Bayes as the zero-frequency problem: a count of 0 gives an estimated probability of exactly 0.

A Prior's Peak: MAP

Keep the prior, and still take only the peak — the **maximum a posteriori** estimate. With a $\mathrm{Beta}(a, b)$ prior, the same derivative gives

$$\hat{\theta}_{\mathrm{MAP}} = \arg\max_{\theta} p(D \mid \theta) p(\theta) = \frac{h + a - 1}{n + a + b - 2}$$

prior	data	MLE h/n	MAP	posterior mean
$\mathrm{Beta}(1, 1)$ flat	3 of 3	1	1	0.80
$\mathrm{Beta}(2, 2)$	3 of 3	1	**0.80**	0.71
$\mathrm{Beta}(2, 2)$	8 of 10 (Lecture 02)	0.80	**0.75**	0.71

The prior shifts the peak as if $a - 1$ yeses and $b - 1$ noes had been seen before the data — a **flat** prior shifts nothing. MAP pulls 3 of 3 back to 0.8 — and still reports one number.

Speaker notes

Derivation: the log-posterior is, up to a constant, $(h + a - 1)\log\theta + (n - h + b - 1)\log(1 - \theta)$, the log-likelihood of a dataset with $a - 1$ extra successes and $b - 1$ extra failures. Its maximum is therefore $(h + a - 1)/(n + a + b - 2)$, valid when both exponents $h + a - 1$ and $n - h + b - 1$ are non-negative and not both zero (for $a, b < 1$ and few data the posterior density is unbounded at the boundary). Beta(1, 1) adds nothing and MAP equals MLE.

Rows of the table. Beta(2, 2) with 3 of 3 gives posterior Beta(5, 2): mode $4/5 = 0.80$, mean $5/7 = 0.71$. Beta(2, 2) with 8 of 10 gives Beta(10, 4): mode $9/12 = 0.75$, mean $10/14 = 0.71$; this is the mini-exercise of Lecture 02.

The posterior mean $(a + h)/(a + b + n)$ is a different point from the MAP $(a + h - 1)/(a + b + n - 2)$; they coincide only for a symmetric posterior ($a + h = b + n - h$). MAP is not invariant under reparametrisation (the mode of a density changes when θ is transformed), the MLE is. Penalised fitting of weights, such as ridge regression or weight decay, is MAP with a Gaussian prior (Lecture 04).

Why Everyone Keeps the Peak

	the whole posterior	only the peak
needs	the normaliser $p(D)$ — an integral over θ	a derivative
in 1 dimension	a Beta table	h/n
in a million	an integral nobody can compute exactly	gradient descent
costs	computation	the **spread**

With **1,000 users** the posterior is $\mathrm{Beta}(301, 701)$: standard deviation 0.015. The peak is nearly everything there is to know.

Speaker notes

The full posterior needs $p(D) = \int p(D \mid \theta) p(\theta)\, d\theta$. In one dimension with a conjugate prior this is a known Beta function; in general it is an integral over all parameters, which has no closed form for most models and grows intractable with dimension. The peak needs only the unnormalised log-posterior and its gradient.

Numbers for the note: Beta(301, 701) has standard deviation $\sqrt{ab/((a+b)^2(a+b+1))} = 0.0145$ and central 95% interval $[0.272, 0.329]$ (Lecture 02). With n observations the posterior standard deviation shrinks roughly as $\sqrt{\theta(1-\theta)/n}$, so with enough data the posterior concentrates around the peak and the peak is an adequate summary.

The approximation fails where the data say little: small samples, many parameters relative to data, and predictions far from the observed inputs. The last case persists even with large datasets; Bayesian linear regression (Lecture 07) shows predictive uncertainty growing away from the data, which a point estimate of the weights cannot express.

Mini-Exercise

Two analysts both report $\hat{\theta} = 0.3$. One saw **3 of 10** users return, the other **300 of 1,000**.

What does the reported number hide — and which estimate would you bet on?

Speaker notes

Answer: the point estimate hides the spread. 3 of 10 with a flat prior gives Beta(4, 8), 95% interval $[0.11, 0.61]$, standard deviation 0.13; 300 of 1,000 gives Beta(301, 701), interval $[0.27, 0.33]$, standard deviation 0.015. The interval widths are 0.50 and 0.057, a ratio of about 9, close to the factor $\sqrt{1000/10} = 10$ expected from a spread shrinking as $1/\sqrt{n}$.

The two estimates agree in value but not in reliability, so the 1,000-user estimate is the one to rely on. The number 0.3 alone cannot distinguish the two cases; reporting n or an interval alongside the estimate restores the missing information.

Mini-Exercise

Two analysts both report $\hat{\theta} = 0.3$. One saw **3 of 10** users return, the other **300 of 1,000**.

What does the reported number hide — and which estimate would you bet on?

The **spread**. The posteriors of Lecture 02: $[0.11, 0.61]$ after 10 users, $[0.27, 0.33]$ after 1,000 — the same peak, an interval nine times wider. Bet on the second; a point estimate cannot tell you to.

Takeaway

Shortcut 1 — keep the peak. Maximum likelihood (flat prior) and MAP (with a prior) keep the posterior's most probable point. It is cheap — a derivative instead of an integral — and harmless with plenty of data. It drops the **spread**: how sure we are about the parameter.

Generative Classifiers

Guiding Question: A classifier needs $P(\text{fraud} \mid x)$. Bayes' theorem builds it from how fraud *looks* — what would it take to model that, and why does nobody do it in full?

Two Routes to the Same Posterior

A labelled table of transactions supports **both** routes to $P(\text{fraud} \mid x)$, where x is the vector of red flags:

DISCRIMINATIVE — LECTURE 04

Model $p(y \mid x)$ directly

A score for each x, as logistic regression will give (Lecture 04). Nothing is said about how x arises.

GENERATIVE — TODAY

Model $p(y)$ and $p(x \mid y)$

How does a *fraudulent* transaction look, and how an honest one? Then convert with Bayes.

$$P(y \mid x) = \frac{p(x \mid y) \cdot P(y)}{\sum_{y'} p(x \mid y') P(y')}$$

Speaker notes

This is Bayes' theorem of Lecture 02 with the hypothesis replaced by a class y and the evidence by a whole attribute vector x; the colour slots are the same (posterior, likelihood, prior, normaliser). The denominator is the law of total probability over the classes, $p(x) = \sum_{y'} p(x \mid y')P(y')$. For classification only the arg max over y matters, so the denominator can be ignored for decisions; it is needed for calibrated probabilities.

Both routes are estimable from the same labelled table. The discriminative route fits $p(y \mid x)$ directly, as logistic regression does (Lecture 04); the generative route fits $P(y)$ and $p(x \mid y)$ and derives $P(y \mid x)$. The generative route is a modelling choice, not a necessity imposed by the data. It models more than classification needs, namely the full joint $p(x, y) = p(x \mid y)P(y)$.

Why Model How Fraud Looks?

A discriminative model — logistic regression (Lecture 04) — gives $P(\text{fraud} \mid x)$ directly. What can a model of $p(x \mid y)$ do that it cannot?

Speaker notes

All three capabilities follow from having the joint $p(x, y)$.

Explicit prior: the posterior odds are the prior odds times the likelihood ratio, so a change of base rate (1 in 1,000 to 1 in 200) only replaces $P(y)$ while $p(x \mid y)$ stays valid, provided the class-conditional distributions do not change between the settings. A discriminative model absorbs the training base rate into its parameters and must be corrected afterwards.

Missing attributes: $p(x_{\text{obs}} \mid y) = \sum_{x_{\text{mis}}} p(x_{\text{obs}}, x_{\text{mis}} \mid y)$ by marginalisation; under the naive assumption the missing factor simply drops out of the product. A model of $p(y \mid x)$ has no principled way to do this without a model of x.

Novelty: $p(x) = \sum_y p(x \mid y)P(y)$ measures how typical a transaction is. A small $p(x)$ flags an input unlike the training data (outlier or distribution shift), which $p(y \mid x)$ alone cannot detect because it is normalised for every x.

The cost is a full distribution over all attributes per class. These three points are compared concretely in Lecture 04.

Why Model How Fraud Looks?

A discriminative model — logistic regression (Lecture 04) — gives $P(\text{fraud} \mid x)$ directly. What can a model of $p(x \mid y)$ do that it cannot?

- ✓ **Keep the prior explicit.** The 1 in 1,000 of Lecture 02 is a separate factor: a new market with 1 in 200 changes $P(y)$, not the whole model.
- ✓ **Handle a missing field.** An attribute nobody observed can be summed out of $p(x \mid y)$.
- ✓ **Notice the strange.** A transaction with tiny $p(x)$ under *both* classes is unlike anything seen — a score alone cannot say so.

> The price: $p(x \mid y)$ is a **joint distribution over all attributes**, one per class.

The Joint Explodes

30

yes/no red flags per transaction

→

$2^{30} - 1$

≈ 1.07 billion numbers for $p(x \mid \text{fraud})$ alone

→

100

frauds among 100,000 transactions to estimate them

Almost every cell of the table is **empty**. Its count is 0, so its maximum likelihood estimate is 0 — every fraud that looks slightly new becomes "impossible".

Speaker notes

A joint distribution over 30 binary variables has $2^{30} = 1{,}073{,}741{,}824$ cells; since they sum to 1, $2^{30} - 1 = 1{,}073{,}741{,}823$ are free parameters, per class. With 100 fraud cases at most 100 cells have a non-zero count, so at least 99.99999% of the cells have MLE exactly 0, and any fraud pattern not seen in training receives $p(x \mid \text{fraud}) = 0$ and therefore $P(\text{fraud} \mid x) = 0$ (or an undefined $0/0$ if the pattern was not seen among honest transactions either).

This is the failure of the MLE on little data seen with three heads in three flips in Lecture 03, multiplied across a billion cells. More data does not solve it: filling the table reliably would need far more frauds than cells. The remedy is a structural assumption that reduces the number of parameters, such as conditional independence (naive Bayes) or a sparse graph of dependences (Bayesian networks).

Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

Speaker notes

Each class needs a full distribution over the $2^3 = 8$ flag patterns, which has $8 - 1 = 7$ free numbers because the probabilities sum to 1. Two classes give 14; the class prior $P(\text{fraud})$ adds one more ($P(\text{honest}) = 1 - P(\text{fraud})$ is determined). Total: 15. Typical miscounts are 16 (counting both class probabilities) and 17 (counting all 8 cells per class).

General formula for K classes and p binary attributes: $K(2^p - 1) + (K - 1)$. Under the naive assumption it becomes $Kp + (K - 1)$, for the example $2 \cdot 3 + 1 = 7$.

Mini-Exercise

A fraud model uses only **three** yes/no flags.

How many numbers does the generative model need — both classes and the class prior together?

15. Each class needs a full table over $2^3 = 8$ flag patterns: 7 free numbers (they sum to 1). Two classes give 14, and $P(\text{fraud})$ adds one.

Takeaway

The generative route models **how the data arises** — $p(y)$ and $p(x \mid y)$ — and converts with Bayes. It keeps the prior explicit and can answer questions about x itself, but it must pay for a **joint over all attributes**, which explodes.

Naive Bayes

Guiding Question: Its central assumption is almost always false, and it classifies well anyway. How can both be true?

Conditional Independence

Definition. A and B are **conditionally independent given** C, written $A \perp\!\!\!\perp B \mid C$, iff

$$P(A \cap B \mid C) = P(A \mid C) \cdot P(B \mid C) \quad \Longleftrightarrow \quad P(A \mid B \cap C) = P(A \mid C)$$

Reading: **once C is known, B tells us nothing more about A.**

Among frauds only: does a new device make a foreign IP more likely?

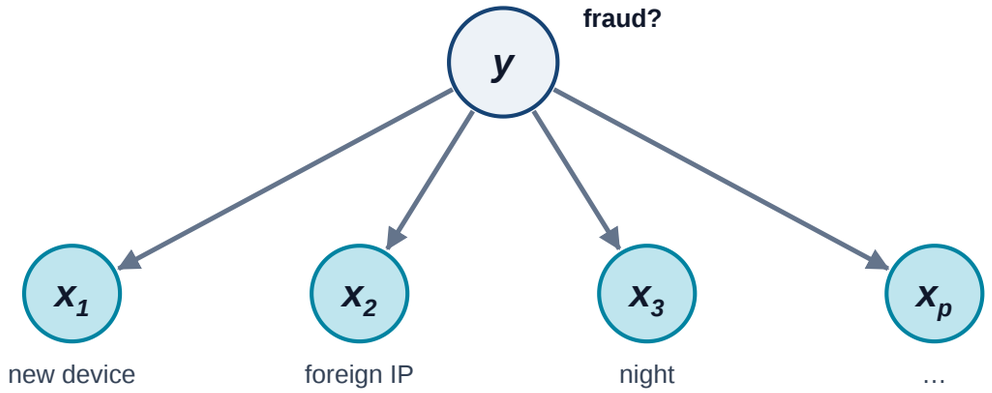
Speaker notes

The two forms are equivalent whenever $P(B \cap C) > 0$: dividing $P(A \cap B \mid C) = P(A \mid C)P(B \mid C)$ by $P(B \mid C)$ gives $P(A \mid B \cap C) = P(A \mid C)$. The relation is symmetric in A and B. For random variables it must hold for every combination of values, and the condition is evaluated separately within each value of C.

Conditional and marginal independence are different properties; neither implies the other. Two flags can be dependent overall because both are more frequent among frauds, yet independent within each class; conversely, conditioning can create a dependence between independent causes (Lecture 05 treats both cases with a counted example).

Answer to the question: in an account take-over the attacker typically uses both a new device and a foreign IP, so among frauds the two flags are positively dependent. With the figures used later in this module, $P(F \mid D, \text{fraud}) = 0.55/0.60 \approx 0.92$, compared with $P(F \mid \text{fraud}) = 0.70$. This is the dependence that naive Bayes assumes away.

The Naive Assumption



$p(y) \cdot p(x_1 \mid y) \cdot p(x_2 \mid y) \cdot \ldots \cdot p(x_p \mid y)$

Assume the attributes are **independent given the class**:

$$p(x_1, \ldots, x_p \mid y) = \prod_{j=1}^{p} p(x_j \mid y)$$

flags	full joint, per class	naive
3	7	3
30	1,073,741,823	30

Shortcut 2 — assume it apart. Dropped: every dependence *within* a class.

Speaker notes

In the graph the class y points to every attribute and no arrow joins two attributes. Read as a Bayesian network (Lecture 05), each missing arrow is an independence claim: every attribute is independent of all others given the class. The attributes are still dependent marginally, through the class.

Parameter count: with the assumption, $p(x \mid y)$ needs one Bernoulli parameter per binary flag and class, p numbers instead of $2^p - 1$. For a categorical attribute with k_j values the count is $k_j - 1$ per class. The three-flag example drops from 15 numbers to $2 \cdot 3 + 1 = 7$; for 30 flags from about $2.1 \cdot 10^9$ to 61.

The model is still generative, since it specifies $p(y)$ and $p(x \mid y)$ and can generate transactions. What it drops is every dependence between attributes within a class (shortcut 2 of the course roadmap). The assumption is almost never exactly true; whether it matters depends on the task, as the last slides of the module show.

Fitting Is Counting

$$\hat{P}(y) = \frac{n(y)}{n} \qquad \hat{P}(x_j = v \mid y) = \frac{n(x_j = v,\ y)}{n(y)}$$

Lecture 02

Each attribute, within one class, is a **categorical** variable — for a yes/no flag, a Bernoulli.

Lecture 03

Its maximum likelihood estimate is the **relative frequency** — the same derivation as h/n for the coin.

Naive Bayes is fitted in one pass over the table, with no optimiser — and every number it uses is a **peak**. Shortcut 1 is taken as well.

Speaker notes

Derivation. Under the naive assumption the log-likelihood of a labelled table is $\sum_i \log P(y_i) + \sum_i \sum_j \log p(x_{ij} \mid y_i)$. Grouping terms by attribute j and class y gives one term $\sum_v n(x_j = v, y) \log \theta_{jvy}$ per attribute and class, plus one term for the class prior. No parameter appears in two terms, so each is maximised separately, and each is the log-likelihood of a categorical distribution whose MLE is the relative frequency (derived for two values as h/n in Lecture 03; for k values with a Lagrange multiplier). The categorical distribution was introduced in Lecture 02 as the distribution for a nominal attribute.

The independence assumption is what makes the MLE a closed-form count: with a full joint the parameters are per combination of values, and the counts would be of entire rows. Logistic regression, by contrast, has no closed-form MLE and is fitted iteratively. Every estimated number in naive Bayes is a peak, so shortcut 1 is taken in addition to shortcut 2.

The Classifier

$$P(y \mid x_1, \ldots, x_p) = \frac{P(y) \prod_j p(x_j \mid y)}{\sum_{y'} P(y') \prod_j p(x_j \mid y')}$$

For two classes, divide the two posteriors — the normaliser cancels:

$$\underbrace{\frac{P(\text{fraud} \mid x)}{P(\text{honest} \mid x)}}_{\text{posterior odds}} = \underbrace{\frac{P(\text{fraud})}{P(\text{honest})}}_{\text{prior odds}} \cdot \prod_{j=1}^{p} \underbrace{\frac{p(x_j \mid \text{fraud})}{p(x_j \mid \text{honest})}}_{\text{likelihood ratio } \mathrm{LR}_j}$$

> Every observed attribute **multiplies the odds** by its likelihood ratio: above 1 it speaks for fraud, below 1 against.

Speaker notes

The first line is Bayes' theorem with the naive product in place of $p(x \mid y)$; the denominator is the law of total probability. In the ratio of two posteriors the denominator cancels, which gives the odds form: posterior odds = prior odds × product of likelihood ratios. Odds and their relation to probabilities, $P = \text{odds}/(1 + \text{odds})$, were introduced with Bayes' theorem in Lecture 02.

Each attribute contributes one factor LR_j. A ratio above 1 raises the odds of fraud, below 1 lowers them, and exactly 1 means the attribute is uninformative. Because multiplication is commutative, the order in which evidence arrives does not matter. An observed absent flag also contributes a factor, $(1 - p_{j,\text{fraud}})/(1 - p_{j,\text{honest}})$, which is usually below 1 for a red flag. For more than two classes, odds relative to one reference class play the same role.

Three Red Flags

100,000 transactions, 100 of them fraud: prior odds $1 : 999$.

flag	among frauds	among honest	likelihood ratio
new device	60%	10%	6
foreign IP	70%	5%	14
night	50%	20%	2.5

A transaction shows all three flags. Is it more likely fraud than not?

Speaker notes

Arithmetic: $0.6/0.1 = 6$, $0.7/0.05 = 14$, $0.5/0.2 = 2.5$; the product is 210, and the posterior odds are $210 : 999$, so $P(\text{fraud} \mid x) = 210/(210 + 999) = 0.174$.

The answer is no: the transaction is still more likely honest. A combined likelihood ratio of 210 is strong evidence, but the prior odds of 1 : 999 are stronger; the posterior exceeds 50% only when the combined ratio exceeds 999. This is the base-rate effect of Lecture 02. Intuitive judgements that call such a transaction "clearly fraud" neglect the base rate.

The model in this example contains exactly these three flags, all observed present, so only the three ratios enter. In a model with more flags, every flag observed absent contributes its own factor; for night that factor would be $(1 - 0.5)/(1 - 0.2) = 0.625$.

Three Red Flags

100,000 transactions, 100 of them fraud: prior odds $1 : 999$.

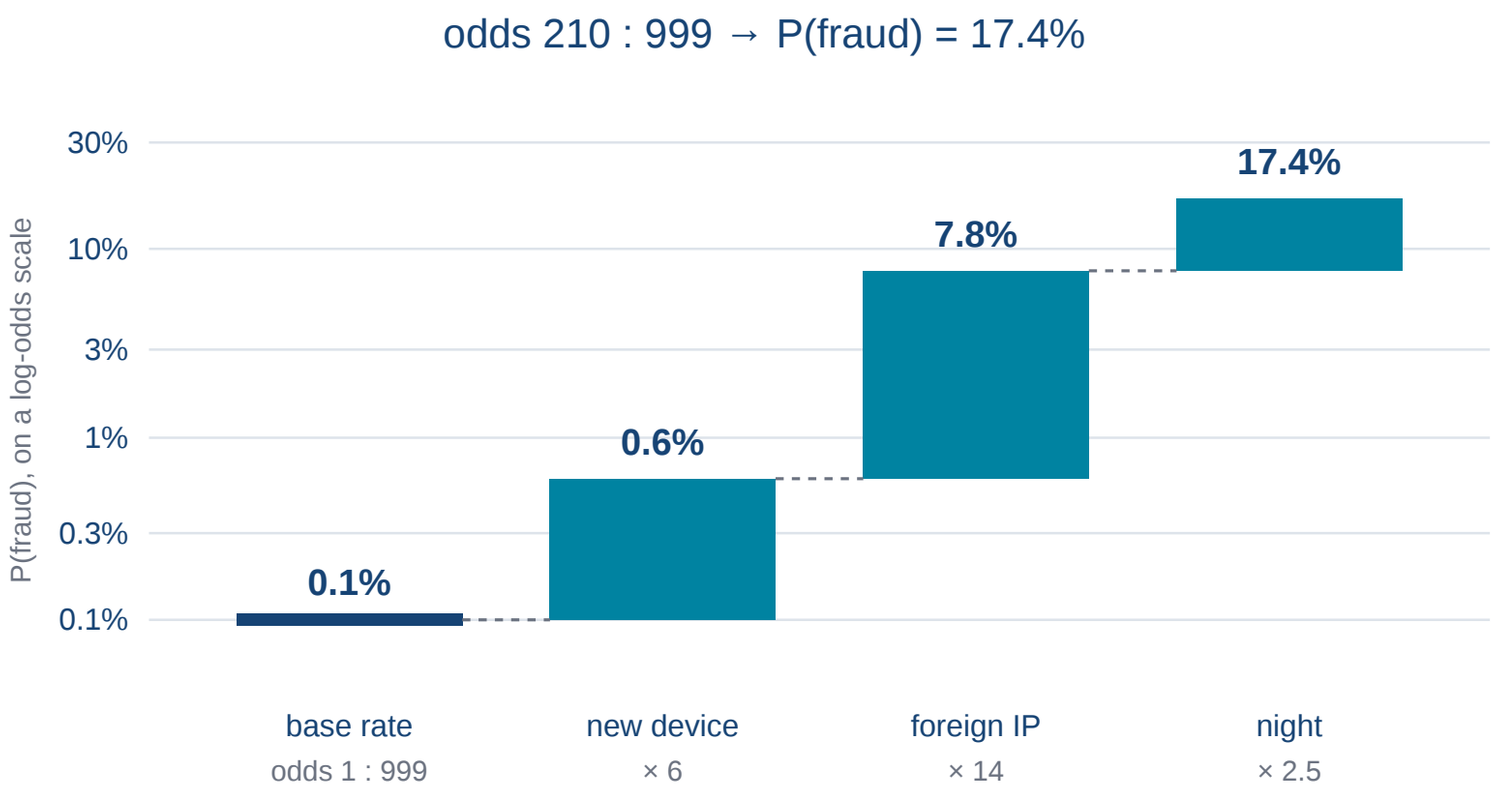
flag	among frauds	among honest	likelihood ratio
new device	60%	10%	6
foreign IP	70%	5%	14
night	50%	20%	2.5

A transaction shows all three flags. Is it more likely fraud than not?

$$\text{odds} = \frac{1}{999} \cdot 6 \cdot 14 \cdot 2.5 = \frac{210}{999} \quad \Longrightarrow \quad P(\text{fraud} \mid x) = \frac{210}{1{,}209} \approx 17\%$$

> Three red flags, each one strong — and still **mostly honest**. The base rate of Lecture 02, once more.

Evidence Adds Up



Predict before each click: which flag moves the probability most?

Speaker notes

On a log-odds axis each likelihood ratio is a step of fixed height $\log \mathrm{LR}_j$, independent of the current position: the log-odds are the sum of the log prior odds and the log likelihood ratios. Natural logarithms: prior $\log(1/999) = -6.91$; steps $\log 6 = 1.79$, $\log 14 = 2.64$, $\log 2.5 = 0.92$.

Step 1 (new device): odds $6/999$, probability 0.6%. Step 2 (foreign IP): odds $84/999$, probability 7.8%. Step 3 (night): odds $210/999$, probability 17.4%. The foreign IP has the largest ratio and therefore the largest step; night is the weakest of the three, although it moves the probability more in absolute terms than the first flag did, because probability is not linear in log-odds.

A sum of per-attribute contributions plus a constant is a linear score. For binary attributes the naive Bayes log-odds are exactly linear in x, the same functional form as logistic regression (Lecture 04); the two differ in how the weights are estimated (Lecture 04).

One Zero Vetoes Everything

A new flag: **payment to a crypto exchange**. Among the 100 frauds it never occurred.

$$\hat{P}(\text{crypto} \mid \text{fraud}) = \frac{0}{100} = 0$$

A transaction with this flag gets posterior odds **exactly 0** — whatever the other thirty flags say. One unseen value vetoes all other evidence.

KEEP THE PEAK, FAILS

"Not seen in 100 frauds" became "impossible for a fraud" — the three heads in three flips of the maximum-likelihood module, in a single cell.

Speaker notes

This is the zero-frequency problem. The MLE of a count of 0 is the probability 0, and a single factor of 0 makes the whole product 0, so the posterior odds of fraud are 0 regardless of all other likelihood ratios. If a value has count 0 in both classes, the posterior is the undefined $0/0$.

The cause is the same as with three heads in three flips in Lecture 03: the MLE on a small count is an extreme value. With 100 frauds, any flag with a true rate below about 1% is likely to have count 0 (for a true rate of 1% the probability of no occurrence in 100 is $0.99^{100} = 0.37$). In a log-odds implementation the zero appears as $\log 0 = -\infty$. Standard repairs add pseudo-counts (next slide) or replace zeros by a small floor value; the first has a probabilistic justification.

Smoothing Is a Prior

Laplace smoothing adds one pseudo-count to every value of attribute j (k_j values):

$$\hat{P}(x_j = v \mid y) = \frac{n(x_j = v,\ y) + 1}{n(y) + k_j}$$

For a yes/no flag ($k_j = 2$) this is $\frac{h+1}{n+2}$ — the **posterior mean** $\frac{a+h}{a+b+n}$ of Lecture 02 under the flat prior $\mathrm{Beta}(1,1)$. For $k_j > 2$ values, the flat Dirichlet does the same.

count among 100 frauds	MLE (peak)	smoothed (mean)
0	0	$1/102 \approx 0.010$
60	0.60	$61/102 \approx 0.598$

> Under the flat prior the peak of 0 of 100 is still 0. Smoothing quietly **leaves the peak for the mean** — exactly where the peak was wrong.

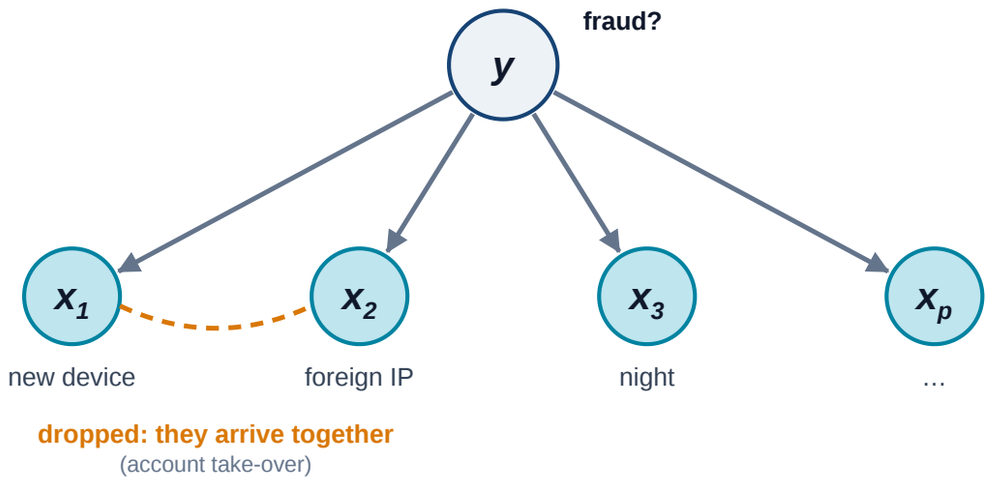
Speaker notes

Derivation: with a Beta(1, 1) prior on $\theta = P(\text{flag} \mid y)$ and h occurrences among $n(y)$ cases, the posterior is Beta$(1 + h, 1 + n - h)$ with mean $(h + 1)/(n + 2)$, which is Laplace's formula for $k_j = 2$. The Dirichlet distribution is the generalisation of the Beta to k values; under the flat Dirichlet$(1, \ldots, 1)$ the posterior mean is $(n_v + 1)/(n + k)$. The same posterior mean after 3 of 10 users was $4/12 = 0.33$ in Lecture 02.

Under the flat prior the posterior mode is still h/n, so the zero survives if only the peak is kept. Laplace smoothing switches from the peak to the mean. On well-populated cells the switch is negligible (0.60 vs 0.598); on empty cells it replaces 0 by $1/102 \approx 0.0098$.

Add-α smoothing, $(h + \alpha)/(n + 2\alpha)$, is the posterior mean under Beta(α, α), equivalent to 2α imagined observations. Larger α is a stronger prior towards one half, like the "near a half" prior of Lecture 02; $\alpha = 0.5$ is known as Jeffreys' prior for the Bernoulli.

Counting Evidence Twice



In an account take-over, a new device and a foreign IP **arrive together**. Suppose both appear on 55% of frauds and 4% of honest transactions:

	ratio for the pair	$P(\text{fraud})$ with night
naive Bayes	$6 \cdot 14 = 84$	17.4%
the real pair	$0.55/0.04 \approx 14$	3.3%

Correlated flags are **counted twice**, so naive Bayes is **over-confident**. Its ranking of transactions often survives; its probabilities do not.

Speaker notes

The dashed edge in the figure is the dependence between new device and foreign IP that naive Bayes omits. Naive Bayes multiplies both likelihood ratios as if they were independent pieces of evidence, $6 \cdot 14 = 84$. The pair's actual likelihood ratio is $P(D, F \mid \text{fraud}) / P(D, F \mid \text{honest}) = 0.55 / 0.04 = 13.75$. With night added, the posterior odds are $13.75 \cdot 2.5 / 999 = 34.4 / 999$, so $P(\text{fraud} \mid x) = 3.3\%$ instead of 17.4%. Under independence within the classes the pair would occur on $0.6 \cdot 0.7 = 42\%$ of frauds and $0.1 \cdot 0.05 = 0.5\%$ of honest transactions; the assumed 4% among honest ones is eight times that, which is where the error comes from.

Correlated attributes therefore push naive Bayes posteriors towards 0 and 1: the probabilities are not calibrated (Lecture 04). Classification by arg max depends only on which side of the decision threshold the posterior falls, so it can remain correct even when the probabilities are badly off. Domingos and Pazzani (1997, Machine Learning 29) show that naive Bayes can be optimal under zero–one loss in many cases where the independence assumption is violated, which is why it classifies well despite a false assumption.

Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

Speaker notes

Answer: up. The duplicated flag contributes its likelihood ratio twice, so the odds become $210 \cdot 14/999 = 2940/999$ and $P(\text{fraud} \mid x) = 2940/3939 = 0.746$, from 17%.

The true information content is unchanged, since a copy of an attribute carries no new evidence: the correct likelihood ratio of the pair (flag, copy) is just the ratio of the flag, 14. Naive Bayes cannot detect duplication, because the model has no parameter for the relation between attributes. A duplicate is the limiting case of perfect correlation; strongly correlated attributes act as partial copies. Real feature sets contain them routinely: the same information coded twice, features derived from others, and indicators that are near-functions of each other. Removing redundant attributes, or a model that captures the dependence (a Bayesian network, Lecture 05), avoids the double counting.

Mini-Exercise

A bug feeds the **foreign-IP** flag into the model twice.

Does the fraud probability of our three-flag transaction go up, down, or stay — and what does that say about correlated attributes?

Up — from 17% to about 75%. The odds get a second factor 14: $2{,}940 : 999$. Naive Bayes cannot tell a copy from new evidence; a strongly correlated attribute is almost a copy.

Takeaway

Naive Bayes **assumes the joint apart** — attributes independent given the class — and **keeps the peak** of every count; smoothed, it keeps the Beta posterior *mean* instead. It is cheap, needs little data and ranks well. Its probabilities are over-confident wherever attributes move together.